



INFORMATICS
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Module : DOC 333 – Introduction to programming

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Problem understanding

This game is a simple implementation of the rock-paper-scissors game, with a few modifications. Instead of using rock, paper, and scissors, the game uses three characters named pinky, inky, and polly. The player competes against the computer, with each making a choice between the three characters. The game follows the traditional rules of rock-paper-scissors: pinky beats inky, inky beats polly, and polly beats pinky. The winner of each round is determined based on the comparison of the player's and the computer's choices, with the score being tallied after each round. The game continues until the player decides to end it. At the end of the game, the winner is determined based on who has the higher score. The game is easy to understand and play, making it a popular choice for quick and simple entertainment. This has a few rules, they are ;

- Pinky > Inky
- Inky > Polly
- Polly > Pinky

Player choice	Computer choice	Rule	Output	Else
Pinky	Inky	Pinky > Inky	Computer won	Player won
Inky	Polly	Inky > Polly	Computer won	Player won
Polly	Pinky	Polly > Pinky	Computer won	Player won

Table 1 - Rules

Algorithm

Step 1 – Start

Step 2 – Initialize variable

Step 3 – Open a list of choices with Pinky, Inky & Polly

Step 4 – Initialize the player score to zero

Step 5 – Initialize the computer score to zero

Step 6 - Input name of the player.

Step 7 – Start a loop until the player decides to End the game

Step 8 – If player enters ‘E’, break the loop and end the game

Step 9 – If player enters something other than (Pinky, Inky, Polly), print an error message and continue the loop

Step 10 – Computer chooses a random choice from the list choices (Pinky, Inky, Polly)

Step 11 – if the choice of the computer and player is equal then print, “It’s a Tie”

Step 12 – If player choses Pinky and computer choses Inky, Player wins.

Step 13 – If player choses Inky and Computer choses Polly, Player wins.

Step 14 – If player choses Polly and computer choses Pinky, Player wins.

Step 15 – else computer wins.

Step 16 – If player wins, award 1 point for the player.

Step 17 – If Computer wins, award 1 point for the computer.

Step 18 – After every round of game, display player name with the score & computer name with the score.

Step 19 – Display the winner.

Step 20 – Continue the loop until the player ends the game.

Step 21 – When the player ends the game

Step 22 – Calculate the final score of the computer as well as the player.

Step 23 - When the player ends the game, display the computer name and the final score of the computer.

Step 24 – and display the player name and player score.

Step 25 – If player score is greater than computer score, then display “player wins” (Player name).

Step 26 – If computer score is greater than player score, then display “Computer wins.”

Step 27 – If computer score is equal to player score, then display “Tie”.

Step 28 - End

Python code

```
*Pinky, Inky, Polly Game.py - C:/Users/USER/AppData/Local/Programs/Python/Python311/Pinky, Inky, Polly Game.py (3.11.1)*
File Edit Format Run Options Window Help
1 import random
2
3 #list of valid choices
4 choices = ('pinky', 'inky', 'polly')
5
6 #Initialize
7 player_score = 0
8 computer_score = 0
9 player_name = input("Enter Player name : ")
10
11 # Start a while loop that runs until the player decides to end the game.
12 while True:
13     player_choice = input("Choose pinky, inky, or polly (or 'E' to End the game!): ")
14
15     # If the player enters 'E', End the game
16     if player_choice == 'E':
17         break
18
19     # If the player's choice is not a valid choice, print an error message and continue the loop
20     if player_choice not in choices:
21         print("Invalid choice, Try again.")
22         continue
23
24     # Generate a random choice for the computer
25     computer_choice = random.choice(choices)
26
27     # Rules of the game
28     if player_choice == computer_choice:
29         outcome = "It's a tie!"
30     elif player_choice == 'pinky' and computer_choice == 'inky' or \
31          player_choice == 'inky' and computer_choice == 'polly' or \
32          player_choice == 'polly' and computer_choice == 'pinky':
33         outcome = "You win!"
34         player_score += 1
35     else:
36         outcome = "Computer wins!"
37         computer_score += 1
38
39     # Print the outcome
40     print(player_name, "Chose", player_choice, "Computer Chose", computer_choice)
41     print(outcome)|
42
```

Figure 1 - Python Code (1)

```
42
43 # When the player ends the game, print the final scores and find the winner
44
45 print ("Final score of", player_name, "is", player_score)
46 print ("Final Score of Computer is", computer_score)
47
48 # Award the winner based on who has more points and print the result
49 if player_score > computer_score:
50     print(player_name, "win!")
51 elif computer_score > player_score:
52     print("Computer wins!")
53 else:
54     print("It's a tie!")
55
```

Figure 2 - Python code (2)

Test cases

Test Case Number	Inputs	Expected output	Points	Remark
1	Computer : Pinky Player : Inky	Computer wins	Computer = 1	Test case pass
2	Computer : Inky Player : Polly	Computer wins	Computer = 1	Test case pass
3	Computer : Polly Player : Pinky	Computer wins	Computer = 1	Test case pass
4	Computer : Inky Player : Pinky	Player wins	Player = 1	Test case pass
5	Computer : Polly Player : Inky	Player wins	Player = 1	Test case pass
6	Computer : Pinky Player : Polly	Player wins	Player = 1	Test case pass
7	Computer : Pinky Player : Pinky	Match Tied	0	Test case pass
8	Computer : Inky Player : Inky	Match Tied	0	Test case pass
9	Computer : Polly Player : Polly	Match Tied	0	Test case pass
10	Computer : Polly Player : ABC	Invalid choice, try again	0	Test case pass

Table 2 - Test Cases

Screenshots of test cases

```
>>> = RESTART: C:/Users/USER/AppData/Local/Programs/Python/Python311/Pinky, Inky, Polly Game.py
Enter Player name : Saad
Choose pinky, inky, or polly (or 'E' to End the game!): pinky
Saad Chose pinky Computer Chose polly
Computer wins!
Choose pinky, inky, or polly (or 'E' to End the game!): inky
Saad Chose inky Computer Chose pinky
Computer wins!
Choose pinky, inky, or polly (or 'E' to End the game!): polly
Saad Chose polly Computer Chose inky
Computer wins!
```

Figure 3 - Test case (1)

```
>>> = RESTART: C:/Users/USER/AppData/Local/Programs/Python/Python311/Pinky, Inky, Polly Game.py
Enter Player name : Saad
Choose pinky, inky, or polly (or 'E' to End the game!): ABC
Invalid choice, Try again.
Choose pinky, inky, or polly (or 'E' to End the game!): polly
Saad Chose polly Computer Chose polly
It's a tie!
```

Figure 4 - Test case (2)

```
>>> = RESTART: C:/Users/USER/AppData/Local/Programs/Python/Python311/Pinky, Inky, Polly Game.py
Enter Player name : Saad
Choose pinky, inky, or polly (or 'E' to End the game!): PINKY
Invalid choice, Try again.
Choose pinky, inky, or polly (or 'E' to End the game!): INKY
Invalid choice, Try again.
Choose pinky, inky, or polly (or 'E' to End the game!): POLLY
Invalid choice, Try again.
```

Figure 5 - Test Case (3)

```
>>> = RESTART: C:/Users/USER/AppData/Local/Programs/Python/Python311/Pinky, Inky, Polly Game.py
Enter Player name : Saad
Choose pinky, inky, or polly (or 'E' to End the game!): pinky
Saad Chose pinky Computer Chose inky
You win!
Choose pinky, inky, or polly (or 'E' to End the game!): polly
Saad Chose polly Computer Chose polly
It's a tie!
Choose pinky, inky, or polly (or 'E' to End the game!): E
Final score of Saad is 1
Final Score of Computer is 0
Saad win!
```

Figure 6 - Test Case (4)

```

>>> = RESTART: C:/Users/USER/AppData/Local/Programs/Python/Python311/Pinky, Inky, Polly Game.py
Enter Player name : Saad
Choose pinky, inky, or polly (or 'E' to End the game!): inky
Saad Chose inky Computer Chose polly
You win!
Choose pinky, inky, or polly (or 'E' to End the game!): pinky
Saad Chose pinky Computer Chose pinky
It's a tie!
Choose pinky, inky, or polly (or 'E' to End the game!): polly
Saad Chose polly Computer Chose pinky
You win!
Choose pinky, inky, or polly (or 'E' to End the game!): inky
Saad Chose inky Computer Chose pinky
Computer wins!
Choose pinky, inky, or polly (or 'E' to End the game!): pinky
Saad Chose pinky Computer Chose polly
Computer wins!
Choose pinky, inky, or polly (or 'E' to End the game!): polly
Saad Chose polly Computer Chose polly
It's a tie!
Choose pinky, inky, or polly (or 'E' to End the game!): polly
Saad Chose polly Computer Chose pinky
You win!
Choose pinky, inky, or polly (or 'E' to End the game!): inky
Saad Chose inky Computer Chose pinky
Computer wins!
Choose pinky, inky, or polly (or 'E' to End the game!): E
Final score of Saad is 3
Final Score of Computer is 3
It's a tie!

```

Figure 7 - Test Case (5)